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QUARTERLY EXAM – 2025
BIOLOGY

12th Std Total Marks: 70
Time: 3.00 Hours

PART – I (BIO-BOTANY)
Marks: 35

SECTION – I

Note: i)Answer all the questions. ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answers. 8 x 1 = 8

Answer all the questions.

1. Which of the following sentences are correct?
   1. The offspring exhibit only parental combinations due to incomplete linkage
   2. The linked genes exhibit some crossing over in complete linkage
   3. The separation of two linked genes are possible in incomplete linkage
   4. Crossing over is absent in complete linkage

   (a) 1 and 2 (b) 2 and 3 (c) 3 and 4 (d) 1 and 4
2. Transmitting tissue is found in
   (a) micropylar region of Ovule (b) Pollen tube wall (c) Stylar region of gynoecium (d) Integument
3. EcoRI cleaves DNA at
   (a) AGGGTT (b) GTATATC (c) GAATTC (d) TATACC
4. The dominant epistasis ratio is
   (a) 9:3:3:1 (b) 12:3:1 (c) 9:3:4 (d) 9:6:1
5. The prevention of large scale loss of biological interity
   (a) Biopatent (b) Bioethics (c) Biosaiety (d) Biofuel
6. Orthotropous ovule is found in
   (a) Piperaceae (b) Dicot (c) Cactaceae (d) None of these above
7. Number of chromosomes Present in Sugarcane
   (a) 22 (b) 10 (c) 16 (d) 80
8. The transfer of DNA from agarose gels to nitrocellulose membrane
   (a) Southern blotting (b) Northern blotting (c) Western blotting (d) None of these above

SECTION – II

Note: Answer any four questions. 4 x 2 = 8

9. What are the types of plant tissue culture?
10. What is bio remediation?
11. What is gene mapping?
12. What is meant by cytoplasmic inheritance?
13. Differentiate Grafting and Layering.
14. What is Apospory?

SECTION – III

Note: Answer any three questions. (Question No. 19 is Compulsory). 3 x 3 = 9

15. Differentiate incomplete dominance and co dominance
16. List out the functions of the Tapetum.
17. What is the difference between missence and nonsense mutations?
18. Give an account on Cryopreservation.
19. Draw a diagram pBR322.

SECTION – IV

Note: Answer all questions. 2 x 5 = 10

20. (a) Explain the mechanism of crossing order. **(OR)**
    (b) Explain the basic concepts involved in plant tissue culture.
21. (a) Discuss the steps involved in microsporogenesis. **(OR)**
    (b) Mention the application of biotechnology.

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BIOLOGY – ZOOLOGY.

SECTION – I

Note : i) Answer all the questions. ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer. 8*1=8

1. A mRNA molecule is produced by.
   a) Duplication b) Replication c) Translation d) Transduction
2. Which of the following is correct regarding cyclosporin A.
   i) Anti – fungal ii) Anti – parasitic ii) Anti - inflammatory iv) Anti – viral
   a) i, ii, iv b) i, ii, iii c) i, iii ,iv d) ii, iii, iv
3. Match the pathogen with its disease.
   a) Leishmania donavani - i) Ameobiasis
   b) Wuchereria bancrofti. – ii) Kala – azar
   c) Trypanosoma gambiense. – iii) Sleeping sickness
   d) Entamoeba histolytica. – iv) Filariasis
   a) A-ii, B-iv,C-iii, D-i b) A-ii, B-iv, C-I, D-iii c) A-iii, B-iv, C-ii, D-iv d) A-i, B-iv, C-iii, D-ii
4. Which of the following blood group has no antigen.
   a) A b) B c) AB d) O
5. Select the incorrect action of hormonal contraceptive pills from the following
   a) Incubation of spermatogenesis b) Inhibition of ovulation
   c) Changes in cervical mucus imparing its ability to allow passage and transport of sperms.
   d) Alteration in uterine endometrium to make it unsuitable for implantation.
6. Female uterus can expand ..... times during pregnancy.
   a) 10 b) 15 c) 20 d) 25
7. In which type of parthenogenesis are only females produced.
   a) Arrhenotoky b) Thelytoky c) Amphitoky d) Both a & b
8. The wings of birds and butterflies is an example of
   a) Adaptive radiation b) Convergent evolution c) Divergent radiation d) Variation

SECTION – II

Note : Answer any four of the following questions. 4×2=8

9. Define Atavistic organs with an example?
10. Placenta is an endocrine tissue? Justify.
11. Define Bio-remediation and its types?
12. Differentiate lagging strand and leading strand?
13. Write the preventive measures of STD's?
14. Name the two allosomal aneuploidy in human?

SECTION – III

Note: Answer any three questions. Q.No.19 is Compulsory. 3×3=9

15. What is colostrum? Write it's significance.
16. What are the applications of Karyotyping?
17. Explain the mode of sex determination in honey bees.
18. Mention any three salient features of genetic code.
19. Differentiate relative dating from absolute dating.

SECTION – IV

Note : Answer the following questions. 2×5=10

20. a) Describe the structure of extra embryonic membrane. **(OR)**
    b Darwin finches and Australian marsupials are suitable examples of adaptive radiation – Justify it statement.
21. a) Explain the structure of Immunogloblin with a suitable diagram. **(OR)**
    b Describe criss - cross inheritance with an example.

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